Mostafa Jafarian Abyaneh

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Summary

Ph.D. student in Civil Engineering at the University of Maryland, College Park, advised by Prof. Yunfeng Zhang, working in computational solid mechanics. Research spans constitutive modeling of composite, brittle, and soft materials; nonlinear finite element analysis in Abaqus, Ansys, and PLAXIS 3D; and the use of machine learning and Bayesian optimization to calibrate and accelerate material models. First author of two peer-reviewed articles in ASCE's *International Journal of Geomechanics*, with additional co-authored work in the *ACI Materials Journal*.

Education

University of Maryland, College Park — College Park, MD
Ph.D. in Civil Engineering — *2026 – Present*
- *Advisor: Prof. Yunfeng Zhang, Structural Engineering Laboratory*

University of Connecticut — Storrs, CT
Ph.D. in Civil / Structural Engineering | GPA: 4.0/4.0 — *2024 – 2025*
- *Research: Constitutive modeling of composite and soft materials using finite element analysis and Bayesian optimization*
- *Coursework: Advanced Solid Mechanics, Polymer Properties, Finite Element I, Variable Topic Polymer Science*

Dalhousie University — Halifax, NS, Canada
M.A.Sc. in Civil Engineering | GPA: 3.4/4.0 — *2017 – 2020*
- *Thesis: Numerical modeling of concrete-filled fiber-reinforced polymer piles*
- *Funded by a Natural Sciences and Engineering Research Council of Canada (NSERC) grant*

Sharif University of Technology — Tehran, Iran
M.Sc. in Civil / Structural Engineering | GPA: 16.28/20 — *2014 – 2017*
- *Thesis: Constitutive modeling of brittle materials (polymer concrete and rock) under triaxial compression loading*
- *Coursework: Linear & Nonlinear Finite Element Analysis, Design and Analysis of Composite Materials*

Iran University of Science and Technology — Tehran, Iran
B.Sc. in Civil Engineering | GPA: 15.39/20 — *2010 – 2014*

Research Interests

- Computational Solid Mechanics & Nonlinear Finite Element Analysis
- Constitutive Modeling of Composite, Brittle, and Soft Materials
- Machine Learning and Bayesian Optimization for Material Model Calibration
- Damage Initiation and Evolution in Fiber-Reinforced Composites
- Multiscale Modeling & Molecular Dynamics
- Soil–Structure Interaction and Deep Foundations
- Additive Manufacturing of Engineered Materials

Experience

Doctoral Research Assistant — Sep. 2024 – Dec. 2025
University of Connecticut — *Storrs, CT*
- Performed finite element simulations of the elastic–plastic response of a randomized representative volume element (RVE) of a **CFRP composite**, including damage initiation and evolution.
- Built **Python automation** for FE geometry generation, meshing, and stochastic parameter randomization, allowing many composite configurations and material property variations to be analyzed in one batch.
- Applied **Bayesian statistical methods and optimization algorithms** to improve how material properties are randomized across realizations.
- Surveyed CFRP durability and its constitutive modeling, and additive manufacturing / 3D printing techniques, to position the modeling work.
- Developed foundational **molecular dynamics** capability in LAMMPS for subsequent multiscale materials work.

Data Analyst — Nov. 2020 – Aug. 2024
Freelance — *Fredericton, NB, Canada*

- Delivered data preprocessing and standardization projects, raising database accuracy and analytical reliability for client datasets.
- Analyzed large-scale user datasets and turned the results into actionable recommendations for client business decisions.

Graduate Research Assistant Sep. 2017 – May 2020
Dalhousie University *Halifax, NS, Canada*

- Modeled the **axial behavior of FRP piles** embedded in multi-layered soil in PLAXIS 3D.
- Modeled the **lateral behavior of concrete-filled FRP tube piles** with a MATLAB implementation of nonlinear finite element analysis, validated against full-scale field tests from a highway bridge on Route 40, Virginia; published in the *ASCE International Journal of Geomechanics*.
- Ran an experimental study on the effect of **tire-derived aggregate** on cylindrical concrete specimens under uniaxial compression.

Graduate Research Assistant Sep. 2014 – Jan. 2017
Sharif University of Technology *Tehran, Iran*

- Conducted experimental and numerical investigations of concrete behavior under uniaxial and triaxial compression, covering polymer concrete, ordinary cement concrete, lightweight concrete, and lime-mortar soil.
- Developed a **disturbed state concept** constitutive model for the softening response and volumetric deformation of rock under triaxial loading; published in the *ASCE International Journal of Geomechanics*.
- Contributed to peer review for journal submissions in materials engineering.

Publications

1. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2020) "Numerical Modeling of the Lateral Behavior of Concrete-Filled FRP Tube Piles in Sand." *ASCE International Journal of Geomechanics*, Vol. 20, No. 8, 04020108. doi:10.1061/(ASCE)GM.1943-5622.0001725
2. **Jafarian Abyaneh, M.** and Toufigh, V. (2018) "Softening Behavior and Volumetric Deformation of Rocks." *ASCE International Journal of Geomechanics*, Vol. 18, No. 8, 04018084. doi:10.1061/(ASCE)GM.1943-5622.0001200
3. Toufigh, V., **Jafarian Abyaneh, M.**, and Jafari, K. (2017) "Study of Behavior of Concrete under Axial and Triaxial Compression." *ACI Materials Journal*, Vol. 114, No. 4, pp. 619–629. doi:10.14359/51689716
4. Jafari, K., **Jafarian Abyaneh, M.**, and Toufigh, V. (2017) "Experimental Study of Different Types of Concrete in Uniaxial Compression Test." *International Journal of Civil and Environmental Engineering*, Vol. 10, No. 12, pp. 1647–1651.
5. **Jafarian Abyaneh, M.**, Jafari, K., and Toufigh, V. (2016) "Constitutive Modeling of Different Types of Concrete under Uniaxial Compression." *International Journal of Civil and Environmental Engineering*, Vol. 10, No. 12, pp. 1483–1486.
6. **Jafarian Abyaneh, M.** (2019) "Numerical Modeling of Concrete-Filled Fiber-Reinforced Polymer Piles." Master's thesis, Dalhousie University, Halifax, NS, Canada.

Conference Papers

1. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2018) "Soil–Structure Interaction Modeling of FRP Composite Piles." *Canadian Society for Civil Engineering (CSCE) Annual Conference*, Fredericton, NB, Canada.
2. **Jafarian Abyaneh, M.**, Sadeghian, P., and El Naggar, H. (2018) "Tire-Derived Aggregate Concrete for Bridge Applications." *10th International Conference on Short and Medium Span Bridges (SMSB)*, Canadian Society for Civil Engineering (CSCE), Quebec City, QC, Canada.
3. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2017) "Lateral Behaviour of Concrete-Filled FRP Tube Piles." *GeoOttawa*, Ottawa, ON, Canada.

Honors and Awards

- Natural Sciences and Engineering Research Council of Canada (NSERC) research grant, Dalhousie University (2017–2020)

Certifications

- **Complete TensorFlow 2 and Keras Deep Learning Bootcamp** — Udemy (Aug. 2022)
- **Python for Machine Learning & Data Science Masterclass** — Udemy (Jun. 2022)
- **Improving Deep Neural Networks** — DeepLearning.AI on Coursera (Apr. 2021)
- **Neural Networks and Deep Learning** — DeepLearning.AI on Coursera (Apr. 2021)
- **Machine Learning** — Stanford University on Coursera (Apr. 2021)
- **Python for Data Science and AI** — IBM on Coursera (Sep. 2020)

Technical Skills

- **Programming & Tools:** Python, MATLAB, SQL, LaTeX, Typst, Git, Inkscape, Microsoft Office
- **Finite Element & Simulation:** Abaqus, Ansys, PLAXIS 3D, LAMMPS (molecular dynamics), SAP2000, ETABS, AutoCAD
- **Machine Learning & Data:** TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Bayesian Optimization, Statistical Modeling
- **Methods:** Nonlinear Finite Element Analysis, Constitutive Modeling, Damage Mechanics, Representative Volume Element (RVE) Modeling, Uncertainty Quantification, Materials Testing

Test Scores

- **GRE General** (Aug. 2022) — Total 314: Verbal 152, Quantitative 162, Analytical Writing 3.0